Blockchain Notes - Detailed Hinglish Version

1. Blockchain Kya Hota Hai?

Blockchain ek digital record book hai (jaise ek bahi-khata), jo sabhi transactions ka record rakhta hai - permanent aur tamper-proof way mein.

Isme "Chain of Blocks" hoti hai:

Har block ek box ki tarah hota hai jisme important data hota hai:

- Transactions (e.g. Ram -> Shyam Rs.220)
- Time stamp
- Previous block ka hash
- Current block ka hash

Key Features:

- Decentralized
- Immutable
- Transparent

2. Block ka Structure Kaisa Hota Hai?

Har block ke andar hota hai:

1. Transaction Data
2. Hash
3. Previous Hash
4. Nonce
5. Timestamp

3. Hashing Kya Hai?

Hashing ek mathematical function hai jo koi bhi data ko ek fixed-size code mein convert karta hai.

Example: Rohit -> fihwakffsvbdahj4rnsdaiuhfiusakngfb

- SHA-256 hashing algorithm use hoti hai
- 64 character hexadecimal string generate hoti hai

4. Proof of Work (PoW)

Blockchain Notes - Detailed Hinglish Version

Miners ek nonce dhundte hain jisse hash 00000... se start ho.

Example:

```
for(i = 0; i < infinity; i++) {
    if(hash(i) starts with 00000) {
        break;
    }
}
```

5. Blockchain Working - Example

Transactions:

- Ram -> Shyam Rs.220
- Mohan -> Ram Rs.100
- Ram -> Bank Rs.300

Process:

1. Transactions collect hoti hain
2. Miner unhe block mein daal kar mining karta hai
3. Block add hota hai chain mein

6. Miners Ka Role

Miners naye block banate hain aur unhe reward milta hai (e.g. Bitcoin).

Bitcoin mein har 10 minute mein 1 block add hota hai.

7. Ledger System

Ledger har transaction ka digital record hai. Yeh tamper-proof aur publicly verifiable hota hai.

8. Blockchain vs Bank System

Blockchain: Decentralized, low fees, high transparency

Bank: Centralized, high fees, slow processes

Blockchain Notes - Detailed Hinglish Version

9. Important Concepts

- Digital Product
- Register (User info)
- Nonce (Proof of Work ke liye)
- Corrupt (Data tamper nahi ho sakta)
- Hashcode (Unique ID)

10. Real World Use Cases

1. Cryptocurrency
2. Voting System
3. HealthCare
4. Supply Chain
5. Digital Identity

11. Summary

Blockchain ek digital bahi-khata hai jisme har transaction permanently aur securely store hoti hai. Trust cryptography par based hota hai.

12. Blockchain Diagram (from Excalidraw)

Blockchain Notes - Detailed Hinglish Version

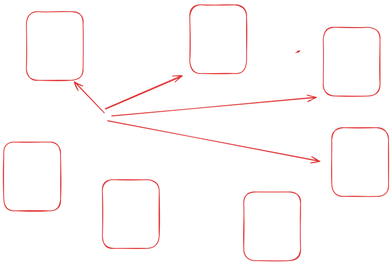
<-- Blockchain -->

DU CA

Chain of Blocks.....

1: Neta log hai... Degree ke upar question? Fake...

1: Register: Mohan kumar



Bank: <--USA-->

Account Digital Product

Register

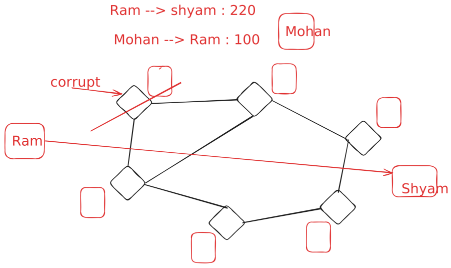
<--Bitcoin--> Banking system replacement ban raha hai

history ko maintain:

~~Balance: 4300~~
4080 4180
transfer
Ram --> shyam : 220

Mohan --> Ram : 100

570
Ledger
Shyam --> Ram: 100
Moahn->Ram:500
Ram-->Rohan: 30
Ram--> Bank--> 300



Miners: Paise

10 min mein 1 block add hota hai

SHA256
32 byte

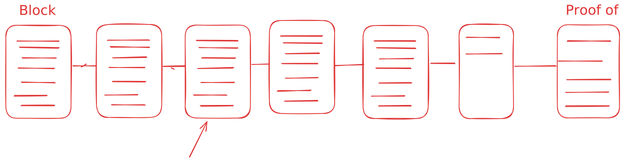
Nonce:
0
00
000
0000
00000

prefix+nonce === 0000(

Number==? Nonce:
number used once
hashCode: 00000xjhxbdgjasf

64 digit

Block



Proof of Work

Number: Hashcode

for(i=0;i<2112030213;i++){
hash(i)

39 0b918943df0962bc7a1824c0555a389347b
4febdc7cf9d1254406d80ce44e3f9

286 00328ce57bbc14b33bd6695bc8eb32cdf2f
b5f3a7d89ec14a42825e15d39df60

hashing

Rohits -----> fihwakffsvbdahj4rnsdaiuhfiusakngfb

Rohit --> fhjkdsafiuhgweiurheiwuqa
Rohan --> 434y3rifasbkjgfbiuasbfdkj

ABC ---> BCD
DEF --> EFG

GHI --> HIJ

fnsdaiuhfiusakngfb → Rohit

BCD ← CDE

fdskhfhqiory498yqowlkerwq